2 Shama Eco-46 404

404

The Grossman model:

The model based on "Human capital Theory" That was published in 1972. Here Grossman Shows the model "How individual allocate their resource to produce health." The model depends on several variables—

* Age: when age increases then health will be depriciates.

* wage: If wealth stock increases the health capital also increases.

* Education: If education increases then optimum health stock increases.

* uncertainty.

→ Comparison between traditional demand and health care demand—

(i) People want health because they demand medicle care inputs to produce health.

(ii) Health is not passively purchased from markets but it is produced in combining time with purchase medicle inputs.

(iii) Health is a capital good, so it does not depriciate instantly.

(iv) Health can be treated both as consumption and investment good.

To Aspect of Health:

* Consumption Aspect: Health makes people help better.

* Investment Aspect: It increases the number of healthy days available to work and to earn income.

* Health investment & home good function

$$I = I(\dot{M}, \dot{T}_H, \dot{E})$$
$$B = B(\dot{X}, \dot{T}_B, \dot{E})$$

Here, T = Total time, 365 day's = T_H (Improving health) + T_B (Producing home goods) + T_L (lost of illness) + T_W (working)

$$T = T_H + T_B + T_L + T_W$$

where,

M = Mediele care.

X = Inputs to Produce home goods.

E = Education (productivity parameter)

I = Health investment

B = Home good function (positive relationship)

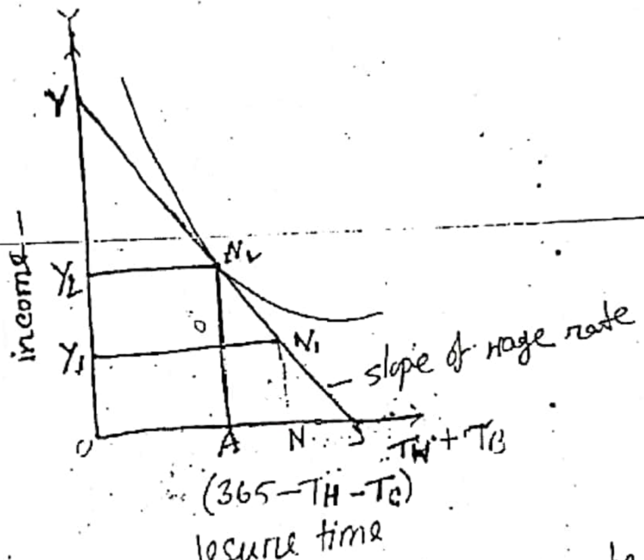
** labor – leisure Trade – offs:

* Here variable T_L and T_H refer to time lost and time spent on healthy ativities, respectively.

* The maximum amount of time that he has available to use either for work – T_W or leisure – T_B. which is – $365 - TH_0 - TL_0$. Thus we can write –

* Time available for work or leisure =

$$365 - TH_0 - TL_0 = T_W + T_B$$



* If Edward choose leisure time OA the he has simiontaneously chosen the amount of time at work AS.

* ~~In point S Edward total~~ amount of time available ~~for either work or~~ leisure time.

* If he only choose point S for the period, he would be ~~choosing to spent~~ all this available time in leisure.

* In Y – axis which represents income, which can be obtained through work.

* This income can be used for market health

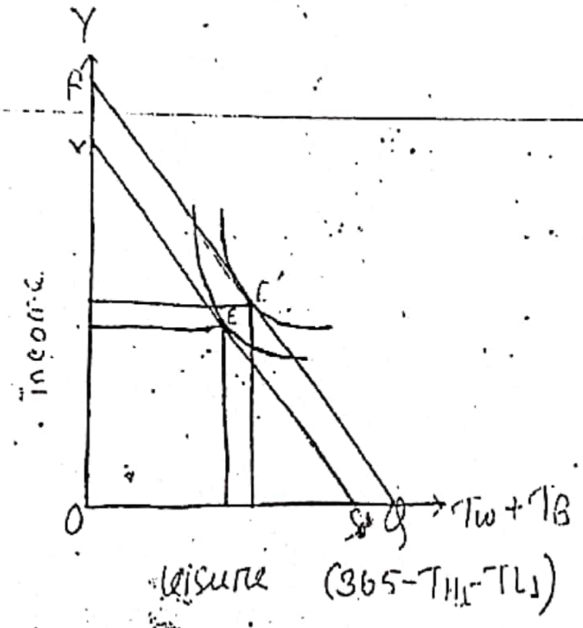
* But if he choosen point S, he will not able to purchase any kind of market goods because he has no wage income.

* If Edward leisure time reduces S from working hour then his income equal to Y_1. which represent his daily wage.

* In economic term, the quantity represents income divided by day's worked.

* which is labor-leisure trade-off.

*** Preference Between leisure and Income:



* In this figure Edward has made a different choice with respect to time spent investing in health status.

* If the time spent on health producing activity is increased from T_{H_1}. T_{H_0}

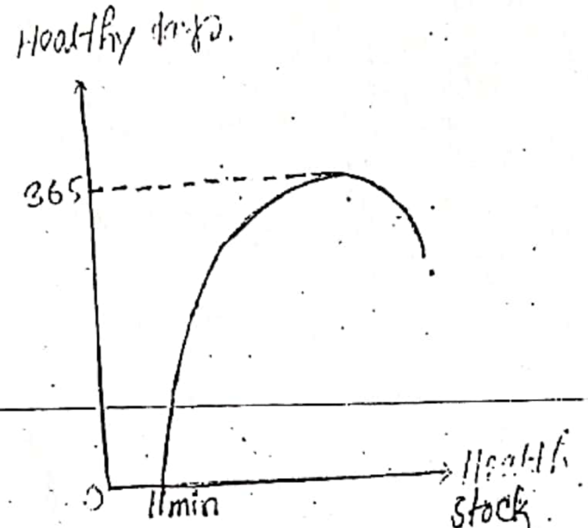
* If the number of days lost to ill, health has been reduced from T_{L_0} to T_{L_1}.

* shifting the income leisure line outward from

can increase his utility moving point from E to E'.

* Not only does investment in health lead to his feeling better, but it also leads to more future income and may lead to more leisure as well.

*** The investment/consumption Aspects of health:



* In this figure, Horizontal axis indicates Health stock and vertical axis indicates Healthy days, which is measured by 365 days.

* The bowed shape of the curve illustrates the law of diminishing marginal returns.

* In Point Hmin which indicates the health stock is minimum and point O indicates the dead period of persons.

* In this curve indicates that, gradually the health stock is increasing but after a certain level the health stock is decreasing day by day the effect of Age.

* So, this is the investment/consumption Aspects of health.

P.T.O

*** Marginal efficiency of investment (MEI) and rate of return.

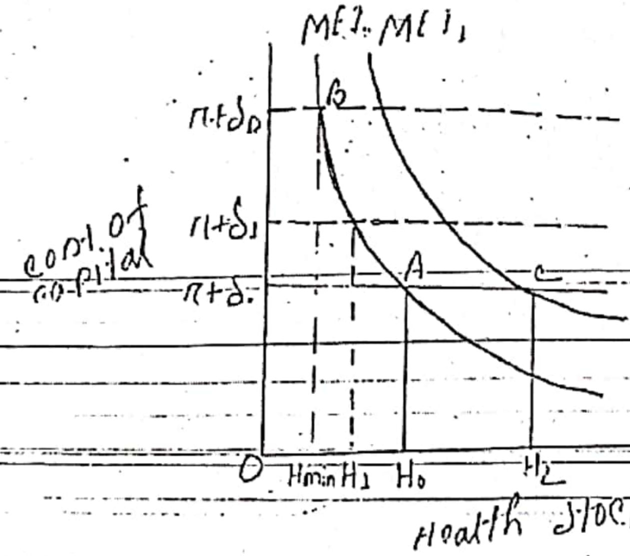
* The MEI can be discribed in terms of the X-ray machine example.

* The first machine purchase and that were return each year $20,000 and the rate of return is $20,000 ÷ $100000 = which is 20% per year.

* The management would buy this machine if the return of the machine is will be greather than the marginal returns.

* If management considerd owning two machines, it would discover that the rate of return on the second X-ray machine would probably be less than first one.

* In this case if the clinic would then purchase the second machine only if it's rate of return were still higher than the interest plus depreciation.



* In the figure MEI describe the patterns of rates of return.

* The cost of capital is shown on the horizontal

* The optimum amount of capital demanded is H_0 which represent the amount of capital. The equilibrium occurs at point A.

* The MEI curve for investements in health also would be downward slopping.

* This occurs because the production function for healthy days exhibits diminishing marginal returns.

* If MEI curve shift's from MEI' to MEI'' then the optimum amount of capital demanded shift's as well as H_0 to H_1, and the cost of capital will remain same ($r+\delta_0$)

*** Changes in Equilibrium (Age, wage, Education and uncertainty)

Age:

* Age is a key-factor for changing health condition. Grossman indicates Hmin is a ~~health~~ optimal health stock and close to zero is a dead of a persons.

* From the begining man gradually grown up, & he demands health care activities.

* At the ~~the~~ level of man's life he will be matured and he buy more health care goods from market to fit his health.

* But increasing age, health will be worse of day by day. in that time people will suffer illness and his wage will be reduces day by day, that's why he can not buy sufficient health care goods from markets.

* In this case, Age is a key factor of reducing health condition and purchase health care goods

* Edward's opti.

* wage:

* wage is effects of a change in level of investment.
* Increase the wage rate as well as the return of healthy days will be increased because he allocate money for buying health care good from market.
* The higher wage imply a higher optimal level of health stock.
* The consumption aspects of health-is that good health makes people feel good.
* we would expect people to reduce their health stock upon retirement.

* Education:

* Education is a special in the demand for health.
* Higher education is most often related to better health and most economist believe that education may improve the efficiency of health and the home care good.
* This explains the widely observed correlation between health status and education.
* Educated people tend to be significantly healthier.
* Educated people are also likely to recognize the benefits of improved health.
* They may enjoy eating nutritious food or doing physical excercise.
* They may recognize the ~~long time~~ dangers of smoking and the long term problems of the over exposure to the sun.

feeling and looking good.

* They may enjoy

*uncertainity:

* The Grossman model starts with the assumption of perfect certainity as to future outcomes but outcomes are not alwayes certain.

* Suppose Edward earnings depend on his health. If he were to fill ill next year, his ability to work and his earnings would be fall and he concerned about that risk.

* An additional doller of investment in health capital this year will help insure against that loss by increasing ~~his~~ his health capital stock and, hence his earnings and consumption next year.

* If Edward increases his future earnings, he would shift his MEI curve outward and he would increase his health capital to reduce his risk.

* This is the uncertainty effect.

—x—

Scanned with CamScanner